

# Uncertainty-aware, explainable skincare recommendation

## Goal

Goal of the project is to build a skincare recommendation app that is:

1. Personalized (Predict what a user may like)  
It should learn what a specific user may like.
2. Explainable  
It should say *why* it recommended something, for example:  
“recommended because it contains ceramides, glycerin, and panthenol, which are common in barrier-repair products.”
3. Uncertainty-aware  
It should know when it is less sure.  
For example:  
“I’m confident about this recommendation” vs “I’m not very sure because this user has very little past history.”

## Problem solved

In skincare,

- there are many products
- each user has tried only a few
- ingredient lists are long and hard to interpret
- reviews are noisy and subjective
- two people with the same skin concern may react differently

Given limited user-product interactions, ingredient information, and product text, can we recommend skincare products accurately, explain the recommendation, and quantify uncertainty?

## Dataset

Sephora Products and Skincare Reviews dataset on Kaggle

## Algorithms used:

### 1. Algorithm 1: Matrix Completion

Suppose we have a giant table: users on one side, products on the other side, ratings inside the cells. But most cells are empty, because each user has tried only a few products. So we have a mostly blank table. Matrix completion tries to fill in the missing cells intelligently.

Limitations: It only learns from the rating table. So if a new product has few or no ratings, matrix completion struggles. This is called the cold-start problem..

### 2. Algorithm 2: Transformer-based product understanding

Example ingredient list:

“Water, Glycerin, Niacinamide, Ceramide NP, Panthenol...”

A Transformer can learn patterns like:

- glycerin + ceramides + panthenol often appear in barrier-repair products
- salicylic acid products often relate to acne/oil control
- fragrance-heavy products may be risky for sensitive users

So instead of only looking at ratings, we also teach the model to understand what the product actually is.

### **3. Algorithm 3: Bayesian Neural Network for uncertainty**

A normal neural network gives one answer:

- “predicted rating = 4.6”

A Bayesian neural network gives:

- “predicted rating is around 4.6”
- “and I am highly sure”
- or
- “predicted rating is around 4.6”
- “but I am not very sure”

So BNN adds uncertainty awareness.

## **METHODOLOGY**

### **1. Build the user-product matrix**

From reviews/ratings:

- rows = users
- columns = products
- entries = ratings

This is the input for matrix completion.

### **2. Run matrix completion**

This fills in missing ratings and gives a baseline recommendation score.

So now we know:

- which products the user may like based on behavior patterns

### **3. Encode product text with Transformer**

Use ingredient lists and review text to build product representations.

So now we know:

- what type of product it is
- what ingredients it contains
- which products are similar in content, not just ratings

### **4. Combine both signals**

For each user-product pair, combine:

- matrix completion score
- product embedding
- maybe metadata like category/price

### **5. Bayesian NN predicts final score + uncertainty**

The Bayesian NN takes all that information and outputs:

- predicted preference score
- uncertainty level

### **6. Recommend top products**

For each user:

- rank products by predicted score
- possibly remove high-uncertainty items if needed
- show explanation using ingredient signals

## **RELEVANT RESEARCH PAPERS:**

Paper 1: Beauty Beyond Words: Explainable Beauty Product Recommendations Using Ingredient-Based Product Attributes (2024)

Paper 2 : Epistemic Uncertainty-aware Recommendation Systems using Bayesian Neural Network, Matrix Factorization, and Deep Ensembles (2025)

These two papers together point to a gap:

- One side focuses on beauty/skincare explainability through ingredients
- The other side focuses on uncertainty-aware recommendation under sparse data

Existing methods either:

- understand products well but do not fully model uncertainty and sparse personalization  
or
- model sparse recommendation uncertainty well but do not understand skincare ingredients and explanations

**Our idea focuses on building a system that focuses on both the methods.**